a quite definite number; but in the world picture of physics it stands for a certain mathematical symbol, which lends itself to manipulation according to quite definite, precise rules. This goes for the height of a tower just as for the duration of the swing of a pendulum or for the brightness of an incandescent lamp. A clear and consistent distinction between the magnitudes of the sense world and the corresponding magnitudes of the world picture is absolutely essential for the clarification of concepts; without such a distinction it is impossible to discuss these questions intelligently and objectively.

It is absolutely untrue, although it is often asserted, that the world picture of physics contains, or may contain, directly observable magnitudes only. On the contrary, directly observable magnitudes are not found at all in the world picture. It contains symbols only. In fact, the world picture even contains constituents which have only a very indirect significance for the sense world, or no significance at all, such as ether waves, partial vibrations, frames of reference, etc. Primarily, such constituents play the part of dead weight or ballast, but they are incorporated because of the decisive advantage assured by the introduction of the world picture—that it permits us to carry through a strict determinism.

Of course, the world picture always remains a mere auxiliary concept. It is self-evident that in the last analysis, the things that really matter, are the occurrences in the sense world and the greatest possible accuracy in predicting them. In classical physics this is achieved as follows: First, an object found in the sense world—for instance, a system of material points—is symbolically represented in some measured condition, i.e. is translated into the world picture. One thus obtains a certain physical structure in a certain initial state. Similarly, suitable symbols are substituted in the framework of the world picture for external influences which operate subsequently on the object. One thus obtains the external forces acting on the structure, and the corresponding boundary conditions. The behavior of the structure is then unambiguously determined for all times by these data, and it can be computed with absolute accuracy from the differential equations of the theory. The coordinates and momenta of all material points of the structure are thus exhibited as quite definite functions of time. If then at any later time the symbols used in the world picture are retranslated into the sense world, one thus obtains a connection between a later occurrence in the sense world and a previous occurrence in the sense world, and this connection can be utilized for an approximate prediction of the later occurrence.

In summary, we may say: While in the sense world the prediction of an occurrence is always associated with a certain element of uncertainty, in the world picture of physics all occurrences follow one another in accordance with precisely definable laws—they are strictly determined causally. Therefore, the introduction of the world picture of physics—and herein lies its